A

Mini Project Report

on

Digital Token Management System

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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CERTIFICATE

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Chapter 1

Introduction

In modern institutional ecosystems, time inefficiency caused by physical queues remains a persistent and underestimated challenge. Whether in educational institutions, healthcare facilities, or government offices, individuals are often required to wait in long lines for routine services such as document verification, fee payments, or administrative approvals. These queues not only reduce operational efficiency but also introduce hidden costs in the form of lost productivity, administrative overhead, and user dissatisfaction.

The concept of "queue anxiety" further highlights the psychological burden experienced by individuals uncertain about waiting times and service progression. Traditional queue management systems lack transparency, flexibility, and scalability, making them unsuitable for high-demand environments.

To address these limitations, Lineless is proposed as a Digital Token Management System that leverages modern technologies to eliminate physical queues. The system enables users to join queues remotely, receive real-time updates, and track their position dynamically. It provides administrators with tools to efficiently manage service flows, ensuring fairness, transparency, and optimized throughput.

This project is designed not merely as an academic exercise but as a scalable FinTech/GovTech-inspired solution capable of transforming service delivery systems across multiple sectors.

1.1 Purpose

The purpose of this document is to provide a comprehensive and structured overview of the Lineless Digital Token Management System, covering its design, functionality, implementation, and overall impact. It is intended to serve as a clear and detailed explanation of how the system works, why it is needed, and how it can be effectively applied in real-world scenarios. The report is designed for academic evaluators and project guides to assess the conceptual understanding and technical execution of the project, for software developers and system designers to understand the architecture and potential improvements, and for institutional administrators who are exploring efficient digital solutions to manage queues and reduce waiting time in service environments.

The primary purpose of this project is to address the common inefficiencies associated with traditional queue management systems, such as long waiting times, overcrowding, lack of transparency, and poor customer experience. In many places like hospitals, banks, and government offices, physical queues often lead to frustration, time wastage, and disorganized service flow. The Lineless Digital Token Management System aims to eliminate these issues by introducing a smart, technology-driven approach where users can generate digital tokens remotely or on-site, track their position in real time, and receive notifications when their turn is approaching. This not only improves operational efficiency but also enhances user convenience and satisfaction.

Furthermore, the system is designed to optimize resource utilization for institutions by enabling better crowd management and reducing manual workload. It allows administrators to monitor token flow, manage service counters, and analyze data for improving service delivery. The purpose also extends to promoting contactless and digital solutions, which are especially relevant in modern environments where hygiene, safety, and efficiency are critical. Overall, this project demonstrates how the integration of technology can transform traditional systems into more organized, scalable, and user-friendly solutions, making it highly relevant for both academic study and practical implementation.

1.2 Problem Statement

The problem addressed in this project arises from the limitations of existing queue management systems, which are mostly manual or partially digital. In many places such as hospitals, banks, and government offices, traditional queuing leads to long waiting times, overcrowding, and confusion among users. These systems lack real-time visibility, making it difficult for users to know their position in the queue or estimate waiting time, which increases frustration and uncertainty.

Additionally, managing multiple service counters becomes inefficient due to the absence of a centralized system, leading to uneven workload distribution and poor service flow. From an administrative point of view, manual processes result in inefficient resource utilization and increased chances of errors. These challenges highlight the need for a smart, scalable, and transparent digital solution that can manage queues efficiently while improving user convenience and overall service quality.

1.3 Objectives

The primary objectives of the Lineless Digital Token Management System are focused on improving efficiency, organization, and user convenience through a structured digital approach. The system aims to provide category-wise queues by using database filter queries, so tokens are arranged separately for categories like Scholarship, Train Pass, or Hostel Issue, ensuring better organization and faster service. It is also designed to generate a unique token for each user request by using Flutter forms for input and auto-increment fields in the database to store and display the token number, which helps maintain accuracy and avoids duplication.

Another important objective is to allow users to cancel their token if they no longer need the service, while only the admin has the authority to mark a token as complete in the database, ensuring proper control and preventing misuse. Additionally, the system focuses on managing admin-defined breaks in queues by using a breaks table in the database and time comparison logic in the backend, so users are notified when the queue is paused and when it resumes. Overall, these objectives aim to create a smart, transparent, and user-friendly system that simplifies queue management and enhances the overall service experience.

1.4 Scope

The scope of the Lineless Digital Token Management System extends across multiple sectors, making it a versatile and impactful solution for modern queue management.

Educational Institutions: In educational environments, the system can be used for managing queues related to scholarships, train concessions, and hostel services, ensuring that students are served in an organized and timely manner. It also helps in streamlining administrative processes by reducing manual workload and improving efficiency in handling student requests.

Healthcare Sector: In the healthcare domain, the system plays a crucial role in managing OPD queues and appointment systems. By digitizing the process, it significantly reduces patient waiting time and minimizes overcrowding, leading to a more comfortable and efficient experience for both patients and medical staff.

Public Sector Offices: For government and public service offices, the system enables digital handling of citizen service queues, making the process more transparent and structured. It helps in improving trust and efficiency in government services by providing clear visibility and fair token distribution.

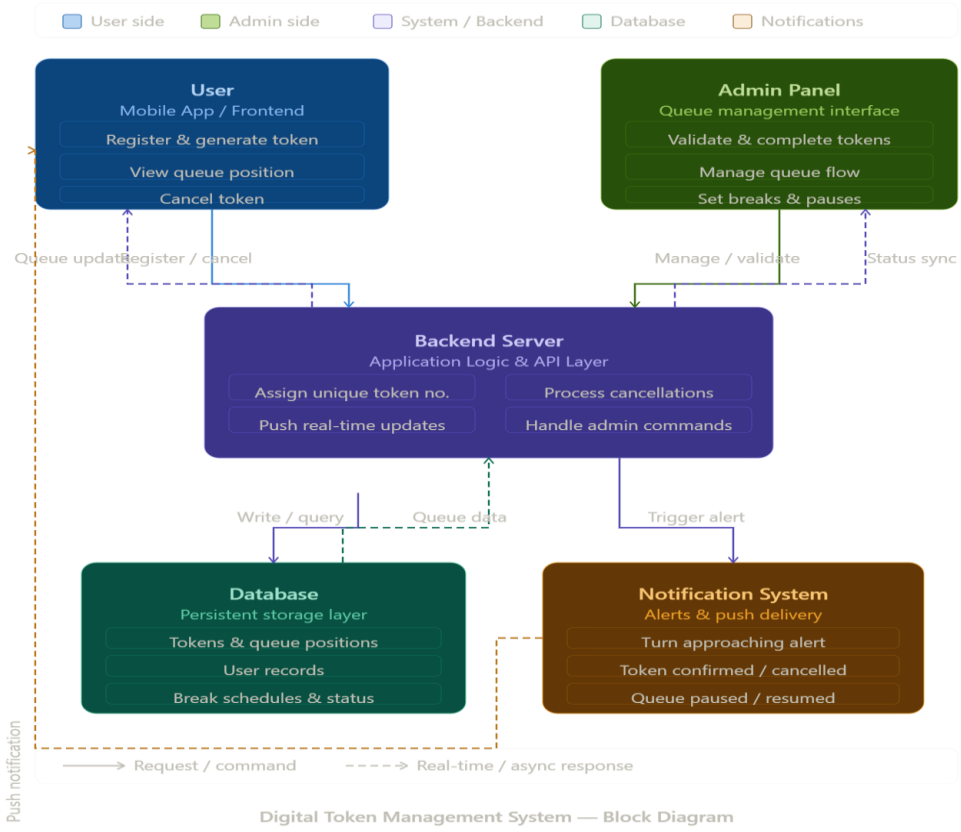
Future Extensions: The scope of the system can be further expanded with advanced features such as integration with AI-based demand prediction to anticipate peak hours, multi-location queue synchronization for handling services across different branches, and analytics-driven decision making to help administrators optimize performance and resource allocation.

Chapter 2

Proposed System

The proposed system aims to introduce a smart, cloud-based digital solution that replaces traditional physical queues with a virtual token management system. Its purpose is to reduce waiting time, prevent overcrowding, and provide users with real-time visibility of their queue status. The system enhances convenience by allowing users to join queues remotely, track their position, and receive timely notifications about their turn.

At the same time, it enables administrators to efficiently manage multiple queues, monitor service progress, and control queue flow in a structured manner. By ensuring transparency, accuracy, and better resource utilization, the system improves overall service efficiency and delivers a more organized and user-friendly queue management experience.



2.1 Features and Functionality

The Lineless Digital Token Management System provides a set of integrated features designed to enhance both user experience and administrative control.

On the user side, the system enables digital token generation through a mobile application, allowing users to join queues remotely without physically standing in line. Users can track their real-time queue position, ensuring transparency and reducing uncertainty. The system also supports token cancellation, giving users flexibility if they no longer require the service. Additionally, push notifications are sent to alert users when their turn is approaching or when there are changes in queue status.

On the admin side, the system offers powerful queue management capabilities through a dedicated Admin Panel. Administrators can create and manage multiple queues, validate tokens, and mark services as completed. They also have control over queue flow, including the ability to pause queues and manage scheduled breaks. The system provides visibility into both pending and completed requests, helping in better decision-making and resource utilization.

At the core, the system follows an algorithmic approach to maintain fairness and efficiency. Each user is assigned a unique token number by the backend server, and queue positions are dynamically updated in real time. When a token is canceled or completed, all subsequent tokens are automatically adjusted, and updates are instantly reflected across the system. This ensures data consistency, fair queue progression, and eliminates the need for manual intervention, making the entire process smooth and highly efficient.

Chapter 3

Project Outcomes

The implementation of Lineless results in:

The implementation of the Lineless Digital Token Management System enhances the overall efficiency of queue handling by introducing a structured and digital approach. The user can generate a digital token through Flutter forms, which is stored in the database with a unique number, ensuring accuracy, proper identification, and elimination of duplication in the system.

The system also provides better organization by allowing the user to view category-wise queues such as Scholarship, Train Pass, and Hostel Issue, which are arranged separately using database filter queries. This categorization helps in reducing confusion, improving clarity, and ensuring faster and more efficient service for different types of requests.

From the administrative perspective, the admin can manage breaks in queues using backend time comparison logic, enabling smooth control over queue flow. During such breaks, users are notified when queues are paused or resumed, which helps in maintaining transparency and keeps users informed about the current status of the system.

Additionally, the system maintains a history of completed tokens for tracking and transparency. This feature allows administrators to monitor past records, analyze performance, and ensure accountability, ultimately contributing to a more reliable and well-managed digital queue system.

Chapter 4

Software Requirements

The implementation of the Lineless Digital Token Management System is supported by a modern and efficient technology stack that ensures smooth performance and cross-platform accessibility. The frontend of the system is developed using Flutter (Cross-platform mobile development), which allows the application to run seamlessly on both Android and iOS devices with a consistent user interface and experience.

On the backend, the system uses Flask (RESTful API framework), which handles the core application logic and manages communication between the frontend and the database. It processes user requests, assigns tokens, manages queue operations, and ensures that all functionalities are executed efficiently through well-structured APIs.

For data storage and additional services, Firebase (Authentication, real-time database, cloud messaging) is used. It provides secure user authentication, real-time database updates for instant queue tracking, and cloud messaging services to send notifications such as token updates and alerts to users. This integration ensures real-time synchronization and reliable data handling across the system.

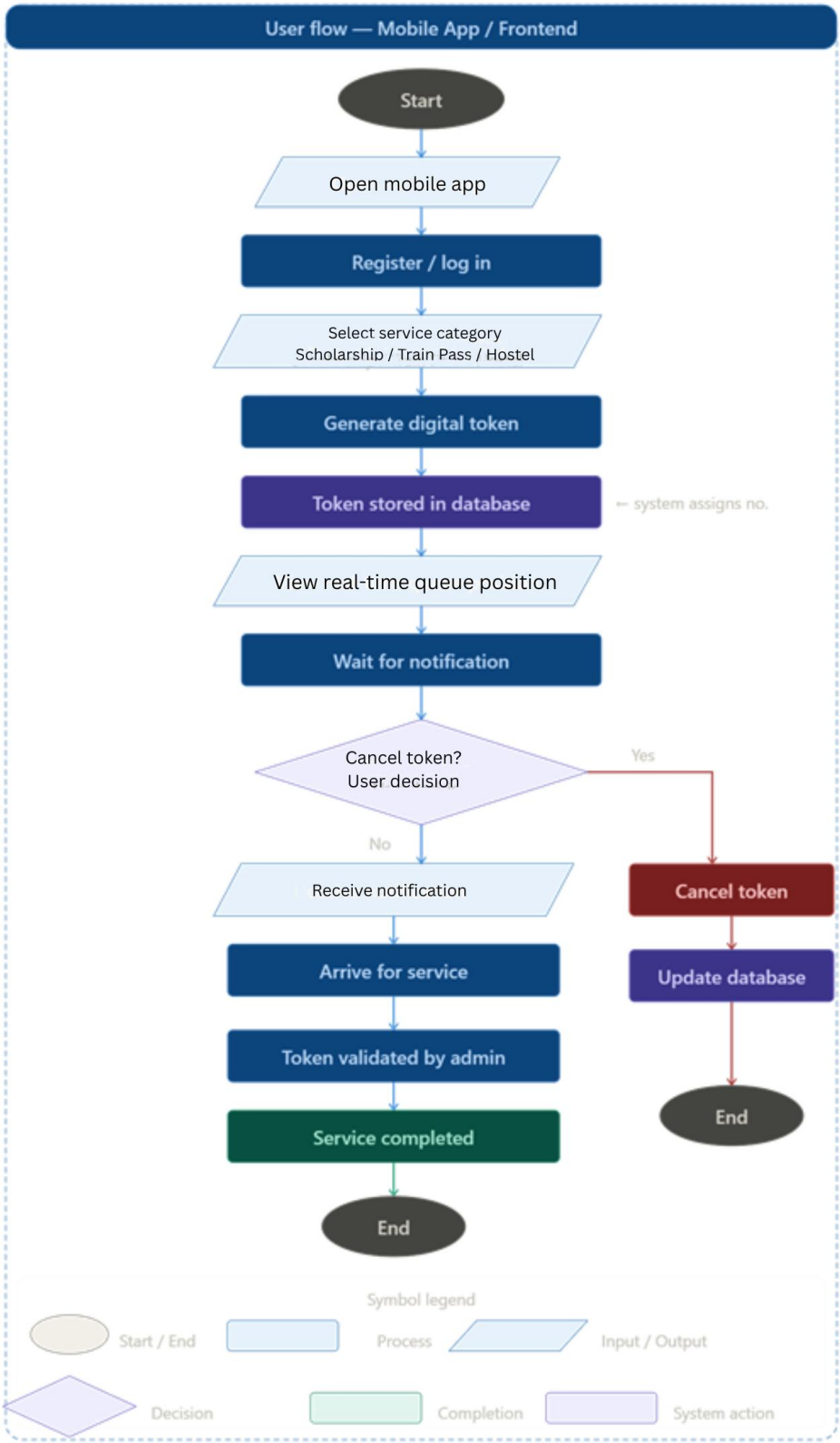
In terms of system requirements, the application requires an Android or iOS device for user access, along with stable internet connectivity to ensure real-time updates and smooth operation. Additionally, a cloud server is required for backend hosting, enabling scalability, remote access, and efficient system performance.

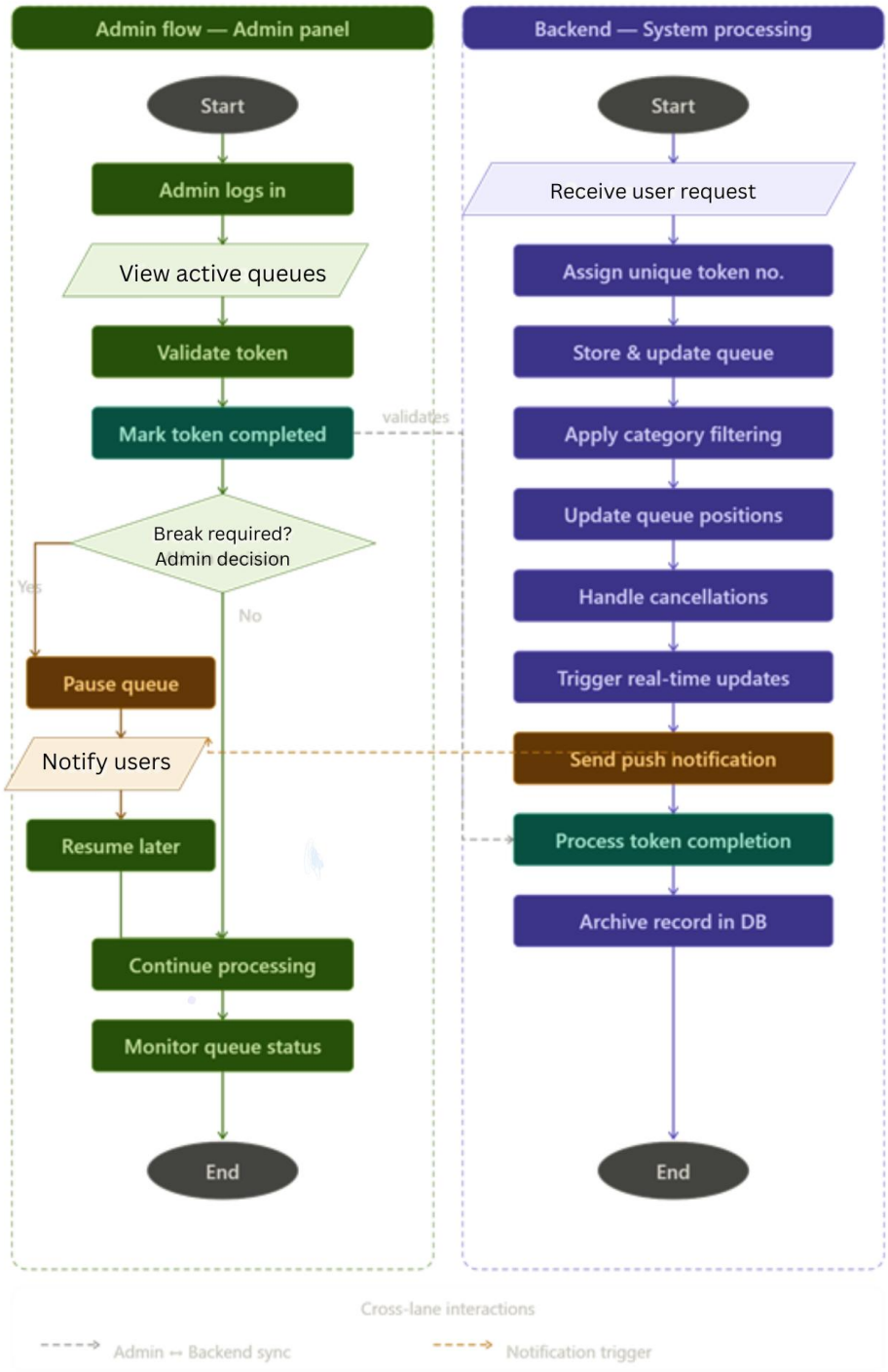
Chapter 5

Project Design

The project design of the Lineless Digital Token Management System focuses on creating a structured, efficient, and scalable solution for managing queues digitally. It is carefully planned to integrate the user interface, admin controls, backend processing, and database management into a unified system that ensures smooth and reliable operation. The design emphasizes simplicity for users while maintaining powerful control and flexibility for administrators.

The system is architected using a modular approach, where each component such as the frontend, backend, database, and notification services performs a specific role while remaining interconnected. This ensures real-time data flow, quick response to user actions, and accurate queue management. Overall, the project design aims to deliver a user-friendly, transparent, and high-performance system that effectively replaces traditional queue methods with a modern digital solution.


User flow — Mobile App / Frontend
Start
Open mobile app
Register / log in
Select service category
Scholarship / Train Pass / Hostel
Generate digital token
Token stored in database
← system assigns no.
View real-time queue position
Wait for notification
Cancel token?
User decision
Yes
No
Receive notification
Cancel token
Arrive for service
Update database
Token validated by admin
End
Service completed
End
Symbol legend
Start / End
Process
Input / Output
Decision
Completion
System action


Admin flow — Admin panel
Start
Admin logs in
View active queues
Validate token
Mark token completed
Break required?
Admin decision
Yes
No
Pause queue
Notify users
Resume later
Continue processing
Monitor queue status
End
validates
Backend — System processing
Start
Receive user request
Assign unique token no.
Store & update queue
Apply category filtering
Update queue positions
Handle cancellations
Trigger real-time updates
Send push notification
Process token completion
Archive record in DB
End
Cross-lane interactions
Admin ↔ Backend sync
Notification trigger

The workflow of the Lineless Digital Token Management System is designed based on a modular architecture that integrates the Frontend Layer, Backend Layer, and Cloud Layer to ensure a smooth and efficient flow of operations. The system is divided into three major workflows: User Workflow, Admin Workflow, and Backend/System Workflow, all of which interact seamlessly to maintain real-time queue management. The Frontend Layer, built using Flutter, provides a responsive and high-performance user interface, allowing users and administrators to interact with the system easily while ensuring a seamless user experience.

In the User Workflow, the process begins when the user opens the mobile application and logs in through secure authentication handled by Firebase. The user then selects a service category such as Scholarship, Train Pass, or Hostel Issue, and generates a digital token through the frontend interface. This request is sent to the Backend Layer, where Flask processes the request and assigns a unique token number. The token details are then stored in the Firebase real-time database, ensuring instant synchronization. The user can view their real-time queue position, as the system continuously updates queue data through the Cloud Layer. If the user decides to cancel the token, the request is processed by the backend, the database is updated, and all subsequent tokens are adjusted automatically. Throughout this process, Firebase Cloud Messaging sends push notifications to inform the user about their turn or any changes in queue status.

In the Admin Workflow, the administrator logs into the system through the frontend interface and accesses the Admin Panel. The admin can view and manage multiple queues simultaneously, as all queue data is fetched from the real-time database. The backend handles admin actions such as validating tokens, marking them as completed, and managing queue flow. When required, the admin can introduce breaks in the queue, which are processed using backend time comparison logic. During these breaks, the system pauses queue progression and sends notifications to users, and once the break ends, the queue resumes automatically. This ensures better control and efficient handling of service operations.

In the Backend/System Workflow, the Flask-based Backend Layer acts as the central processing unit that connects the frontend with the Cloud Layer. It handles all business logic, including token generation, category-wise filtering using database queries, dynamic

queue updates, and processing of cancellations and completions. The Firebase Cloud Layer supports real-time synchronization, secure authentication, and push notifications, ensuring that all updates are instantly reflected across the system. The overall design follows key principles such as scalability to handle multiple users and queues, reliability for consistent performance, low latency for real-time updates, and modular extensibility to allow future enhancements. Together, these workflows create a well-structured, efficient, and intelligent digital queue management system.

Chapter 6

Project Scheduling

The development of the Lineless Digital Token Management System was carefully planned and executed using a structured timeline, as illustrated in Figure 6.1 (Gantt Chart). The Gantt chart provides a clear visual representation of how the project activities were distributed over time, showing the start and end dates of each task along with their sequence and overlap. It highlights how the project was systematically divided into multiple phases such as requirement analysis, system design, frontend and backend development, integration, testing, deployment, and documentation. Each activity is mapped across a timeline, allowing easy tracking of progress and ensuring that all phases were completed within the planned duration.

As observed in Figure 6.1, the scheduling follows a well-organized and partially overlapping structure, which reflects a semi-agile development approach. Certain tasks, especially frontend and backend development, were carried out in parallel to reduce the overall development time and improve efficiency. The chart also shows how dependencies between tasks were managed, ensuring that critical activities were prioritized while maintaining a smooth workflow. Buffer periods were implicitly included between phases to accommodate testing, debugging, and refinement, which helped in maintaining the quality of the system.

The project scheduling approach focused on effective task prioritization, parallel module development, continuous integration, and regular testing. This ensured that issues were identified early and resolved without affecting the overall timeline. The use of a Gantt chart, as shown in Figure 6.1, played a crucial role in coordinating tasks among team members, monitoring deadlines, and maintaining clarity in execution. It served as a roadmap for the entire project, guiding the development process in a structured and disciplined manner.

As a result, the planned scheduling led to the timely completion of all project phases while ensuring efficient resource utilization and smooth coordination among team members. The structured timeline minimized delays and enabled the team to deliver a high-quality,

reliable system. Overall, Figure 6.1 demonstrates how proper planning and scheduling contributed significantly to the successful development of the Lineless Digital Token Management System.

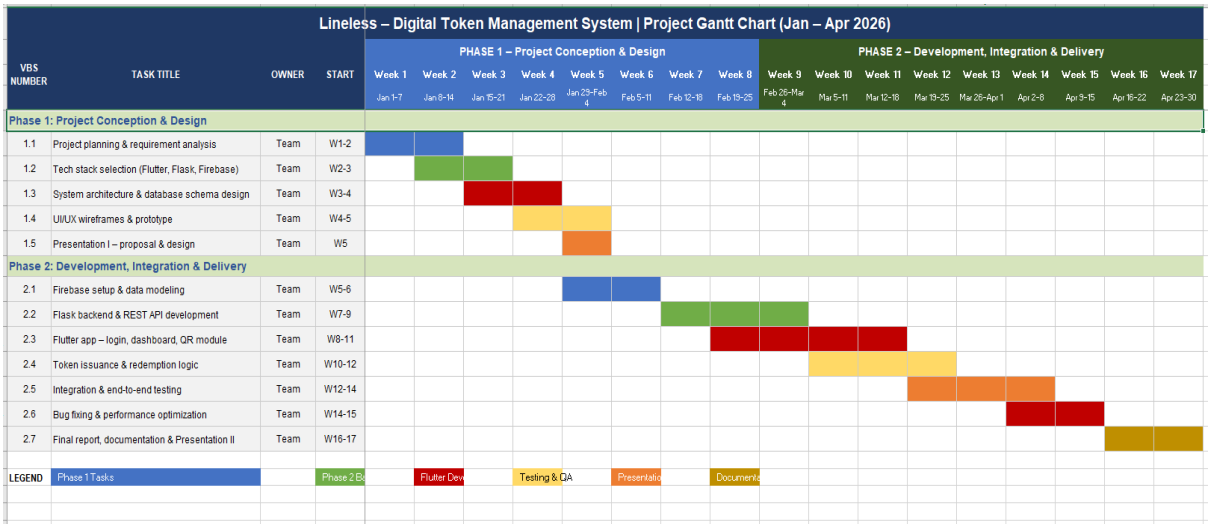


Figure 6.1: Gantt Chart for Lineless Digital Token Management System

Chapter 7

Results

The Lineless Digital Token Management System was successfully implemented and rigorously tested, yielding highly satisfactory results across all major functionalities. The system demonstrated accurate real-time queue updates, ensuring that users could continuously track their position without delays or inconsistencies. It provided a seamless user experience across devices, with the Flutter-based interface maintaining smooth performance and responsiveness on both Android and iOS platforms. Additionally, the system efficiently handled multiple queues simultaneously through category-wise segregation, allowing organized management of services such as Scholarship, Train Pass, and Hostel Issue without any performance degradation. The notification system also proved to be reliable, delivering timely alerts for token generation, queue status updates, and turn notifications, thereby enhancing user convenience.

The practical implementation and working of the system are clearly demonstrated through the application screenshots included in the report, which provide visual evidence of the user interface, token generation process, real-time queue tracking, and admin functionalities. These screenshots validate the successful integration of frontend, backend, and cloud services, showcasing the system's usability and effectiveness in real-world scenarios. Overall, the results confirm that the system operates as a fully functional digital queue ecosystem, capable of being deployed in real-world environments with high efficiency, reliability, and user satisfaction.

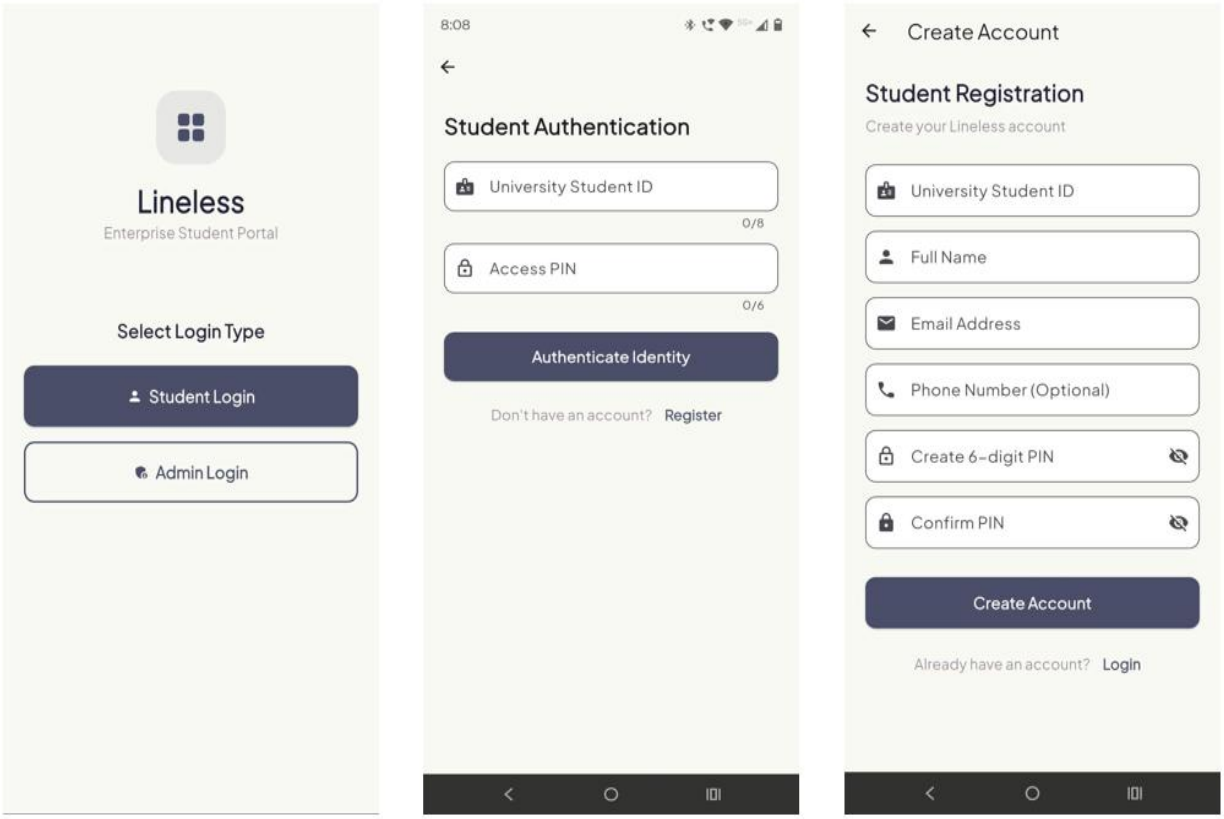


Figure 7.1: User Authentication and Registration Flow

The above figure (Fig 7.1) shows the three main entry points for students and admins in the Lineless portal.

- Login Selection Screen Displays two options: Student Login and Admin Login. Students use this to access token services, while admins use it to manage queues.
- Student Authentication Screen Students verify their identity using their University Student ID and a secure PIN. This ensures that only authorized users can generate tokens and access services.
- Student Registration Screen New users can create an account by entering details such as Student ID, Name, Email, Phone, and a secure PIN. This registration process stores student information for future verification and service requests.

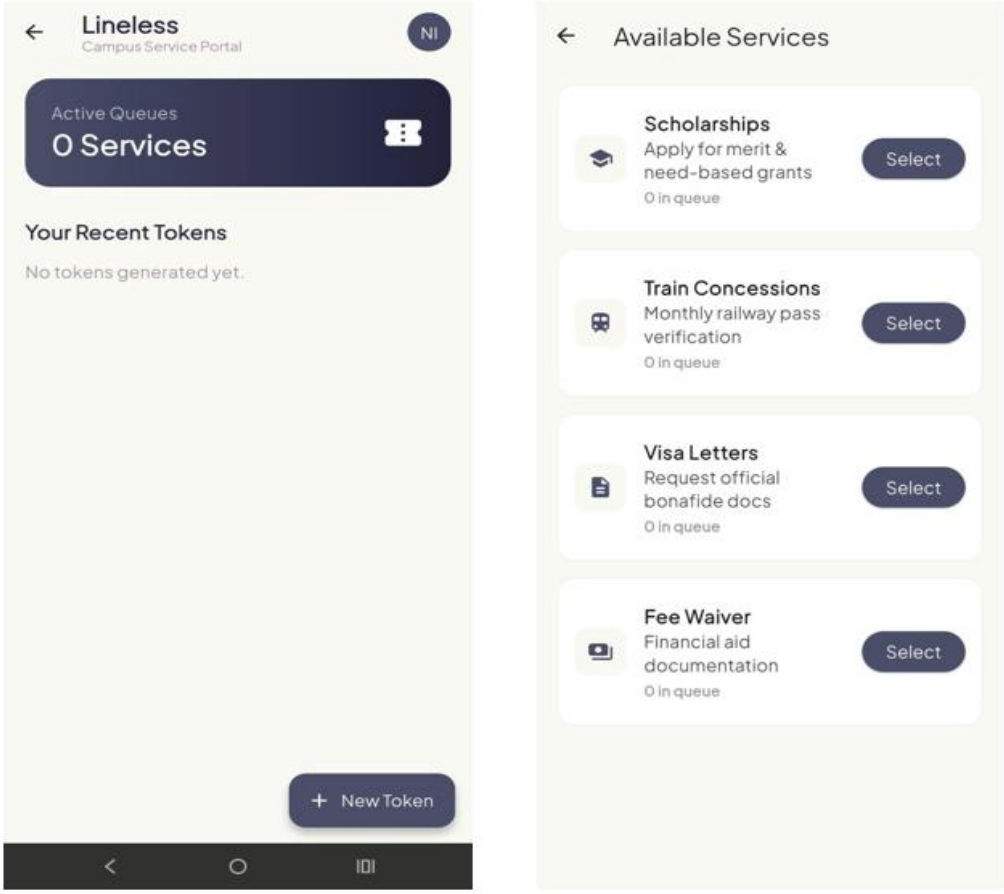


Figure 7.2: Dashboard and Available Services

This figure(Fig 7.2) shows the student's main dashboard and the list of available services in the Lineless portal.

- Active Queues Dashboard Displays the student's active queues and recent tokens. If no tokens are generated, the dashboard shows "No tokens generated yet." A button at the bottom allows the student to create a new token.
- Available Services Screen Shows the different service categories such as Scholarships, Train Concessions, Visa Letters, and Fee Waiver. Each service displays the current queue length and provides a "Select" option for the student to generate a token.

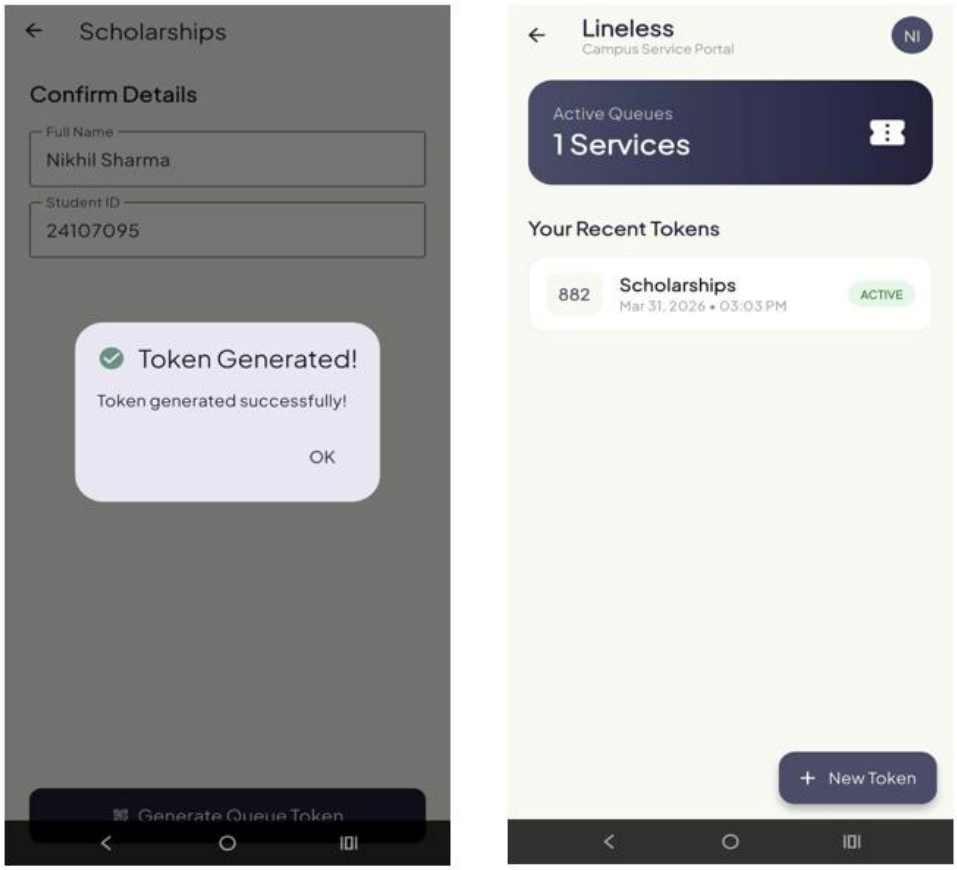


Figure 7.3: Token Generation and Active Token Display

This figure(Fig 7.3) shows the process of generating a token and tracking its status in the Lineless portal.

- Token Generation Screen (Scholarships) The student confirms their details (Name and Student ID) before generating a token. Once submitted, the system displays a confirmation message “Token Generated Successfully,” ensuring the request is recorded in the database.

- Active Token Display Screen After generation, the token appears under “Your Recent Tokens” with details such as token number, category (Scholarships), date, time, and status marked as “ACTIVE.” This helps the student track their position in the queue and know when their turn is approaching.

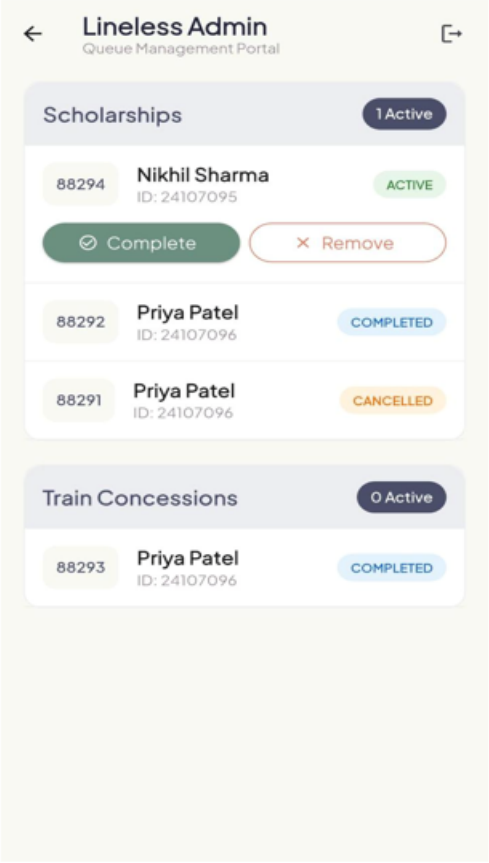


Figure 7.4: Admin Queue Management Portal

This figure(Fig 7.4) shows the administrator's interface for managing student service queues.

- Scholarships Queue Displays tokens with different statuses such as *Active*, *Completed*, and *Cancelled*. Admins have options to mark tokens as complete or remove them if necessary.
- Train Concessions Queue Shows tokens related to train pass verification. Completed requests are recorded here, ensuring transparency and proper tracking of services.

This portal highlights the admin's control over queues, where only admins can finalize completion or cancellation of tokens.

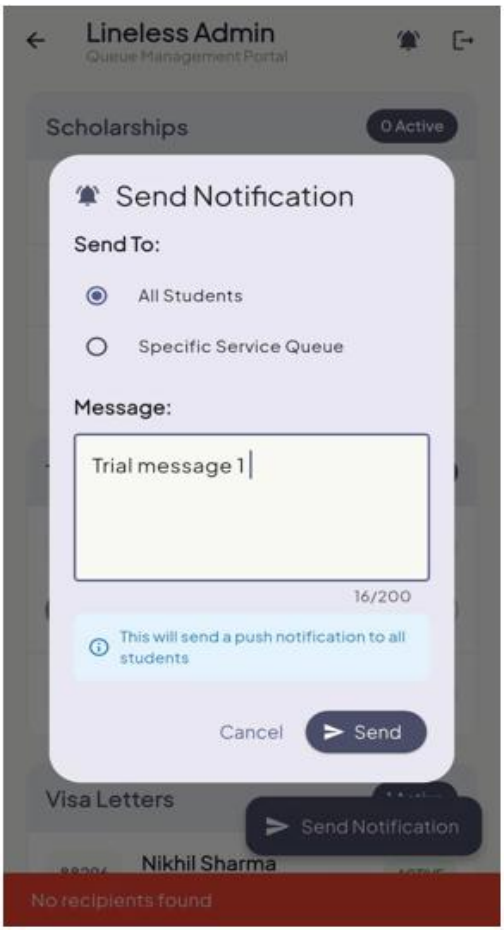


Figure 7.5: Admin Notification Panel

This figure(Fig 7.5) shows the administrator's interface for sending notifications to students.

- Notification Options Admins can choose to send messages either to *all students* or to a specific service queue. This ensures targeted communication depending on the situation.
- Message Input A text box allows the admin to type a custom message (up to 200 characters). This message is then delivered as a push notification to the selected recipients.
- Purpose This feature is useful for informing students about important updates such as queue breaks, delays, or service announcements, ensuring smooth communication between admin and users.

Chapter 8

Conclusion

The Lineless Digital Token Management System successfully addresses the inefficiencies of traditional queue systems by introducing a digital, scalable, and intelligent solution. By eliminating physical waiting lines, the system significantly enhances productivity, improves user experience, and optimizes administrative workflows through better organization and real-time management. Its ability to provide transparent queue tracking, efficient resource utilization, and seamless communication between users and administrators makes it a highly effective modern solution.

Furthermore, with its modular architecture and flexibility, the system can be easily adapted across multiple domains such as educational institutions, healthcare, and public sector services. This adaptability, combined with its cloud-based and real-time capabilities, gives Lineless strong potential to evolve into a large-scale infrastructure solution. In the future, it can contribute to the development of smart cities and support digital governance by enabling more efficient, transparent, and user-friendly public service systems.

References:

[1] Singh, P., "Design and Implementation of Mobile-Based Queue Systems," *Journal of Emerging Technologies in IT*, Vol. 9, Issue 3, pp. 55–62, 2020.

[2] Firebase Documentation, "Firebase Realtime Database and Cloud Messaging," Available: https://firebase.google.com/docs, Accessed: Jan 2026.

[3] Flutter Documentation, "Building Cross-Platform Mobile Applications," Available: https://flutter.dev/docs, Accessed: Jan 2026.

[4] Fielding, R., "Architectural Styles and the Design of Network-based Software Architectures," Doctoral Dissertation, University of California, Irvine, 2000.